

Boise State University - Systems Engineering Paper
Bender Autonomous Robotic Systems Team

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Reason for Systems Engineering

The Bender Autonomous Robotics Team employed systems engineering because it provided a critical structure for managing the complexity of constructing a fully autonomous lunar robot. With over 20 students collaborating across subsystems, and with 3 students leaving after fall and 3 new members joining between January and March, systems engineering practices ensured that knowledge transfer and project continuity were maintained. These practices helped coordinate design decisions, document progress, and clearly define interfaces and requirements so that new members could onboard efficiently without slowing progress. The framework was especially valuable given our compressed timeline after shifting to a spring-focused build schedule, allowing us to set clear objectives and systematically validate performance across the evolving team structure.

Project Management Merit

System Roots

This is Team Bender's first year competing in the NASA Lunabotics Challenge, and our robot is a completely new design. There are no reused components or engineering documentation from prior years. Every element of the system, from chassis to software, was designed, developed, and implemented specifically for this year's competition.

Major Reviews

Team Bender conducted three major design reviews during the development of our autonomous lunar robot: the System Requirements Review (SRR), Preliminary Design Review (PDR), and Critical Design Review (CDR). Each review served as a formal control gate to evaluate design decisions, assess risk, and determine project readiness to progress to the next phase.

System Requirements Review (SRR)

The SRR was conducted internally, without external reviewers. Each subsystem team gave a 10-minute presentation outlining initial design concepts and multiple possible directions, followed by 5 minutes of Q&A from other subteams and project leads. After all presentations, the team held a collaborative design session to review decisions against the Technical Performance Measures (TPMs) and requirements defined in our Project Management Plan (PMP).

Key outcomes included:

- System Design: Initial concepts (e.g., wheels vs tracks, arm vs drum) were explored and compared against regolith, power, and dimension requirements. Chassis focused on ease of implementation, assembly, and mounting for components through the use of an aluminum extrusion skeleton, acrylic top, and screwed-in aluminum side panels.
- TPMs: Several TPMs were clarified or reworded to align more explicitly with testable outcomes. For example, mapping accuracy was refined to “ ± 5 cm” to match sensor expectations.

This review set the foundation for the project's technical direction and confirmed alignment between subsystems before resource commitments.

Preliminary Design Review (PDR)

The PDR was held in front of an external industry review panel including Nardos Ashenafi (Robotics Software Engineer II at Bastian Solutions), Wankun Sirichotiyakul (Robotics Software Engineer III at Bastian Solutions), Makarand Mandolkar (Robotics Software Engineer III at Bastian Solutions), and Camron Collinsworth (Software Engineer at HPE). Each subsystem team gave a 10-minute presentation, followed by a 5-minute Q&A, concluding with a 30-minute open discussion involving all reviewers and the full team.

Reviewer feedback resulted in significant design changes:

- **System Design:** Based on recommendations from the panel, we transitioned from a wheeled drivetrain to tracks to improve terrain traversal, traction, and regolith handling stability.
 - **Before:** The requirement for the drivetrain focused on low-power use and minimized weight with a simple 4-wheel configuration.
 - **After:** New requirement emphasized track durability and modular mounting for ease of replacement and testing.
- **Schedule:** The feedback prompted us to add a dedicated testing phase prior to full system integration, which slightly shifted our final build milestone.
- **TPMs:** We added a TPM for traction and turning radius, and reallocated the expected drivetrain weight and power to account for the more complex track system.
- **Cost Budget:** The transition to tracks also led to a modest increase in drivetrain costs (~\$300), accounted for in our next wave of part orders.

Critical Design Review (CDR)

The CDR was conducted in front of an expanded panel of industry and faculty reviewers, including Nardos Ashenafi (Robotics Software Engineer II at Bastian Solutions), Wankun Sirichotiyakul (Robotics Software Engineer III at Bastian Solutions), Makarand Mandolkar (Robotics Software Engineer III at Bastian Solutions), Akash Junandra (Lead Robotic Systems

Engineer at Bastian Solutions), Kevin Falk (Industrial Designer at PuzL Labs), Aaron Smith (Clinical Assistant Professor at Boise State University), and Conner Patricelli (Clinical Assistant Professor at Boise State University). The team gave a 15-minute overview presentation, followed by 5 minutes of Q&A, and each subsystem then delivered a 10-minute presentation with 5-minute Q&A. The session concluded with a 30-minute open panel discussion.

Key impacts from the CDR included:

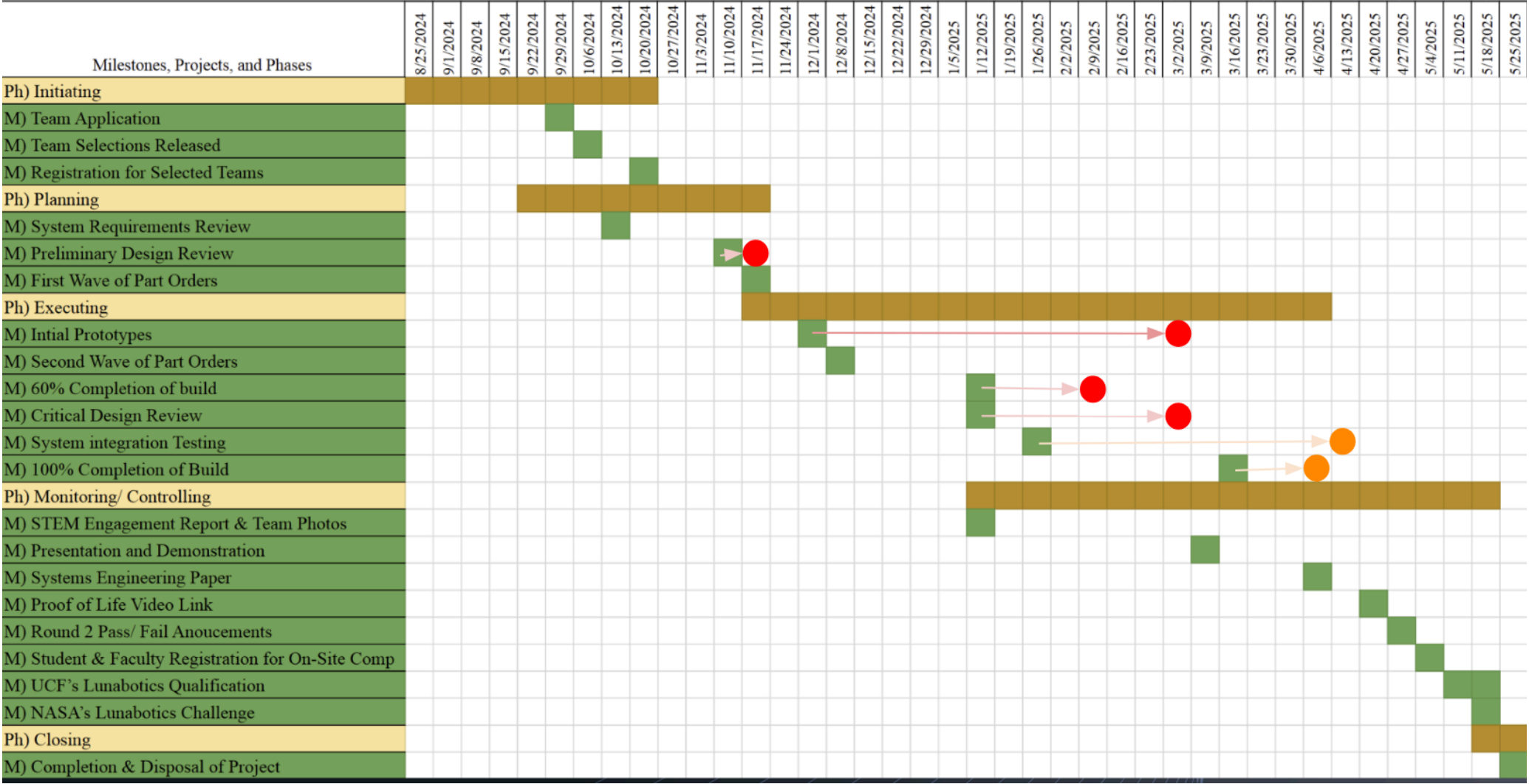
- **System Design:** Reviewers emphasized the need to test the durability of 3D-printed tracks, leading to the inclusion of additional mechanical testing before Bruneau Dunes field trials. Reviewers also noted the computational strain some sensors would induce on the processors, leading to a reevaluation of sensor use.
 - Concerns surrounding weight distribution and center of gravity were also highlighted. These were addressed with careful layout of internal components with emphasis on the position of heavy components like the battery and motors.
- **TPMs:** A TPM for regolith dumping accuracy was added, with a success criterion tied to berm dimensions (1.5m x 0.9m at UCF).
- **Schedule:** The CDR confirmed that our current timeline was realistic, though it reinforced the importance of staying on track with integration testing.
- **Budget:** No major changes were recommended, though we reallocated funds internally to support potential reprints or replacements of track components during final validation.

Overall, the CDR served as a final design checkpoint and confirmed the project's readiness to move into full system testing and deployment.

Schedule of Work

The original project schedule was outlined in our Project Management Plan (PMP) and structured around the NASA Lunabotics competition milestones, with a target completion date in early April 2025. Our plan was divided into the standard project phases: Initiating, Planning, Executing, Monitoring/Controlling, and Closing. Key events were paced according to internal reviews and NASA deliverables.

As shown in the schedule chart, the timeline has evolved significantly throughout the project. Several milestone dates were shifted due to review feedback, delays in subsystem readiness, and adjustments following team member turnover. These changes are marked by red dots (missed or changed milestones) and orange dots (anticipated changes) in the updated Gantt chart.



Schedule Changes

- Preliminary Design Review: Adjusted by a week to accommodate reviewer schedules
- Second Wave Part Order: Adjusted to March to accommodate the team getting back into the project and onboarding new members
- Initial Prototypes: Adjusted due to a slower start to the Spring semester than anticipated.
- 60% Completion of Build: Adjusted due to a slower start to the Spring semester than anticipated.
- Critical Design Review: Adjusted to be before the 100% completion date
- 100% Completion of Build: Adjusted to be after the CDR
- Systems Integration Testing: Adjusted to be after the build completion

Impact on Cost and Technical Requirements

The adjusted schedule did not significantly impact our overall cost budget, as material purchases had been planned across five “waves,” with flexibility built in. However, the extended testing timeline did prompt some reallocation within the budget:

- Funds originally set aside for additional hardware were shifted to support replacement track components and extra consumables for field testing.
- The extended schedule also emphasized the need for refined TPMs. For example:
- The durability and modularity of tracks became a more emphasized requirement.
- TPMs like compute time and mapping accuracy were re-evaluated during testing and simulations.

Schedule as a Project Management Tool

Throughout the project, the schedule was actively used to manage progress, drive weekly meetings, and inform team decision-making. Key milestones were reviewed at the end of each month, and Gantt charts were updated to reflect progress and flag critical paths. When tasks slipped, mitigation steps were implemented, for example, shifting integration deadlines to allow subsystems more development time or reprioritizing tasks to match part arrival. The result is a schedule that remained flexible yet focused, ensuring progress toward a successful full-system test at the Bruneau Sand Dunes and a proof-of-life submission in time for the competition deadline.

Cost Budget

The initial total estimated cost for Team Bender’s lunar excavation system, as submitted in the Project Management Plan (PMP), was \$23,156. This estimate covered all aspects of the project including hardware, electronics, travel, shipping, and student stipends. The budget was

developed with a conservative estimate of travel participation and with early-stage part cost approximations.

As of this writing, our actual expenditures are approximately \$28,691.14, with additional purchases pending final integration testing and travel logistics. A full breakdown of both initial and updated costs is included below, with placeholders for ongoing calculations.

Original Budget Breakdown (PMP)

Category	Subcategory	Estimated Cost
Chassis	Structure, electronics, vision, power	\$8,165.00
Navigation	Mapping, positioning, control, processing	\$0
Regolith Handling	Collection, storage, dumping, electronics	\$1,619.00
Drivetrain	Motors, tracks, mounting, control	\$2,880.00
Travel	Flights, hotel, transportation, shipping, stipend	\$13,372.00
Total	—	\$23,156.00

Note: Subcategories such as Mapping, Planning, Control, etc. were internally budgeted but not itemized as line-item expenses at the time of initial submission.

Budget Status (At Time of Submission)

Category	Estimated Final Cost	Notes
Chassis	\$2001.80	Pending final part purchases
Navigation	\$1308.66	Moved sensors and supporting components from chassis to nav
Regolith Handling	\$2086.00	Slight increase due to custom drum optimization
Drivetrain	\$2480.92	3D-printed track cost is included here
Travel	\$19,880.00	Increased to cover 11 travelers (up from 6)
Miscellaneous	\$933.76	Includes spare parts, shipping increases
Current Spend (to date)	\$8811.14	All current purchases logged and tracked
Estimated Final Total	\$28,691.14	The final figure to be confirmed post-travel

Budget Evolution and Management

Several key changes were made throughout the project:

- **Travel Cost Increase:** Initially budgeted for 6 travelers, the team raised sufficient funds to support 11 students, increasing the travel line item by approximately \$6,500.
- **Hardware Flexibility:** Cost estimates for hardware remained largely stable due to bulk ordering and contingency planning. Slight increases in drivetrain and regolith handling materials were absorbed by a surplus in other categories.
- **Phased Spending Strategy:** The team organized purchasing into five planned “waves” following review gates (SRR, PDR, and CDR) and two prototyping sprints. This allowed us to:
 - Delay large expenses until designs were validated
 - Shift funds based on test results and integration findings
 - Preserve flexibility for unexpected expenses (e.g., track replacements, travel contingencies)
- **Impact on Schedule and Requirements:** No requirement changes resulted directly from cost changes, though additional travel funding enabled more team members to support testing and field validation, improving system performance and safety awareness.

Budget as a Project Management Tool

The budget was reviewed monthly alongside schedule updates. Spending decisions were evaluated against technical objectives, test readiness, and competition deliverables. This alignment between budget and system milestones allowed us to manage risk and respond quickly to reviewer feedback and test outcomes without exceeding our financial constraints.

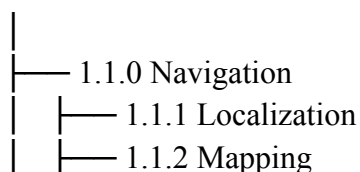
Systems Engineering Merit

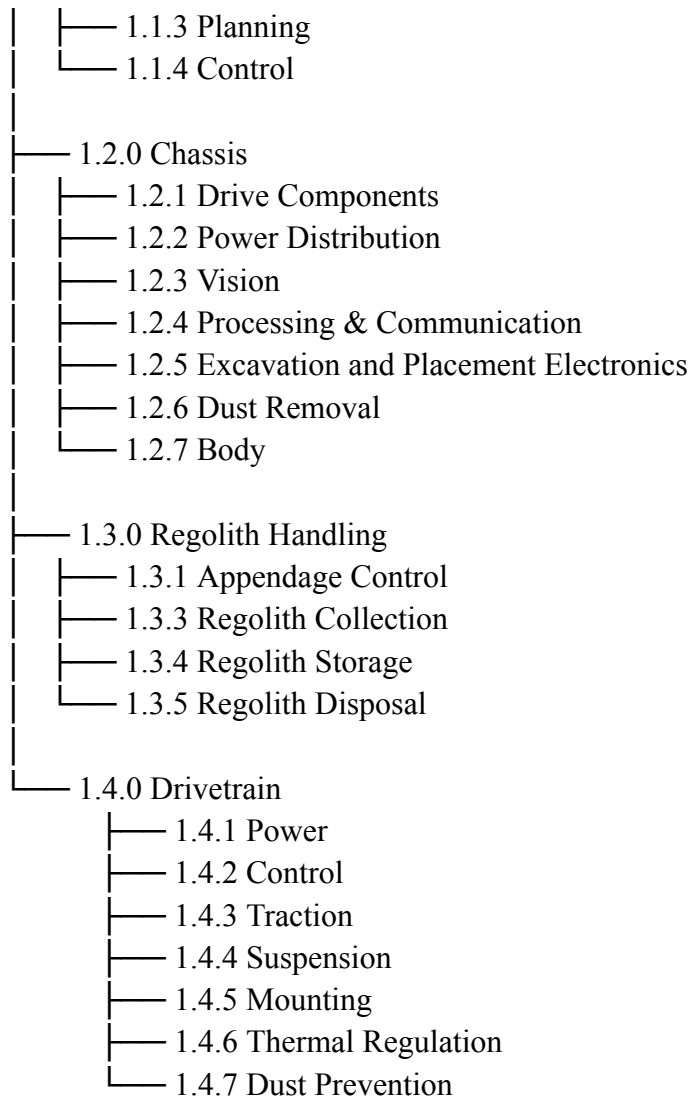
System Hierarchy

The Bender system architecture is organized in a top-down hierarchy, broken into four primary subsystems: Navigation, Chassis, Regolith Handling, and Drivetrain. Each subsystem contains additional layers of components and functions, creating a modular structure for integration and testing. This hierarchy was reviewed and refined at each major control gate, SRR, PDR, and CDR, to guide system decomposition, interface definition, and TPM allocation.

Initial System Hierarchy (Defined in PMP & SRR)

1 - Bender





System Requirements Review (SRR) Focus Areas

During SRR, the team focused on early concept refinement and feasibility analysis of the following hierarchy elements:

- Navigation: Localization (1.1.1) and Planning (1.1.3)
- Chassis: Overall structure (1.2.0), especially the Body (1.2.7)
- Regolith Handling: System integration (1.3.0), with a focus on Collection (1.3.3), Storage (1.3.4), and Disposal (1.3.5)
- Drivetrain: General assembly (1.4.0), traction approach (1.4.3), and mounting design (1.4.5)

These early discussions helped align the design with the defined TPMs in the PMP and led to the first wave of concept validation.

Preliminary Design Review (PDR) Focus Areas

At PDR, the system design had matured and the review focused on deeper subsystem integration and control elements. Focused hierarchy elements included:

- Navigation: Full structure including Localization (1.1.1) and Planning (1.1.3)
- Chassis: Drive Components (1.2.1), Power Distribution (1.2.2), Excavation and Placement Electronics (1.2.5), and Body (1.2.7)
- Regolith Handling: Expanded analysis on Appendage Control (1.3.1), Collection (1.3.3), Storage (1.3.4), and Disposal (1.3.5)
- Drivetrain: Power (1.4.1), Traction (1.4.3), Mounting (1.4.5), and Dust Prevention (1.4.7)

Change at PDR: The Suspension system (1.4.4) was removed from the hierarchy based on feedback from reviewers regarding complexity, weight tradeoffs, slow movement speed, and level nature of the simulated lunar terrain. It was deemed non-essential to achieve our mobility TPMs given the adoption of a track system.

Critical Design Review (CDR) Focus

By CDR, the team had developed and integrated nearly all system components. The review focused on the entire system hierarchy, ensuring compatibility across all modules and assessing readiness for full system testing.

- All elements from Navigation, Chassis, Regolith Handling, and Drivetrain were reviewed.
- Interface definitions, testability of requirements, and modularity were scrutinized.
- Emphasis was placed on ensuring heat dissipation, dust protection, and robustness of subsystem integration—especially for field operations.

The CDR confirmed that the hierarchy was well-scoped, technically sound, and supported full verification and validation planning.

Requirements

The key driving requirements for Team Bender's lunar robot system were derived from the project's primary technical objectives: 1) to maximize autonomous operation, 2) maintain robust mobility across simulated lunar terrain, and 3) effectively collect, transport, and deposit regolith while minimizing dust contamination and system degradation. These requirements span functional design, operations, interfaces, testing, safety, reliability, and energy usage. All requirements derive from the PMP that was previously submitted.

System-Level Requirements

- 1.0.0 – Vehicle Envelope
 - The vehicle shall fit within a volume of 1.5 x 0.75 x 0.75 meters to meet competition constraints.

Navigation Requirements

- 1.1.0 – Navigation Power
 - Navigational computation shall consume no more than 60 Watts at full operation.
- 1.1.1 – Positional Accuracy
 - The navigation system shall determine the vehicle's position relative to a global reference point with an accuracy of ± 5 cm.
- 1.1.2 – Mapping
 - The vehicle shall map the surrounding terrain and detect obstacles with an accuracy of ± 5 cm.
- 1.1.4 – Compute Time
 - The vehicle shall compute terrain data, perform mapping, and determine its position within 500 ms.

Chassis Requirements

- 1.2.0 – Chassis Weight
 - The chassis shall have a mass of 30 kg or less.
- 1.2.2 – Emergency Stop
 - The vehicle shall include a physical, unmodified Commercial Off-The-Shelf (COTS) red e-stop button compliant with OSHA IEC 60204-1, readily accessible to the operator.
- 1.2.7 – Lifting Points
 - The vehicle shall have four lifting points, each equally distributed around the center of mass, with no single point carrying more than 20 kg.
 - Each lifting point shall be clearly marked in accordance with ISO 7000-1368.

Regolith Handling Requirements

- 1.3.0 – Subsystem Weight
 - The regolith handling subsystem shall have an unladen mass of 10 kg or less.
- 1.3.0 – Regolith Performance
 - The vehicle shall excavate, transport, and deposit regolith into a berm with a target area of 1.5m x 0.9m (UCF) and 2.2m x 0.9m (KSC) without significant loss of material.

- 1.3.1 – Power Budget
 - The regolith handling subsystem shall operate using 200 Watts or less.

Drivetrain Requirements

- 1.4.0 – Drivetrain Weight
 - The drivetrain assembly shall weigh 15 kg or less.
- 1.4.1 – Drivetrain Power
 - The drivetrain shall consume no more than 400 Watts at full capacity.
- 1.4.2 – Turning Radius
 - The vehicle shall perform a 180° turn within a radius of 2 meters.
- 1.4.3 – Terrain Traversal
 - The drivetrain shall enable traversal of the simulated lunar terrain while transporting 10 kg of regolith without loss of locomotive capability.

Safety & Interface Requirements

- All external interfaces for power, networking, and control shall be accessible, labeled, and modular to support field maintainability.
- The onboard network shall support USB, Ethernet, and wireless connections through dust-resistant pass-throughs.
- Thermal and dust mitigation measures shall ensure that sensitive electronics operate within safe temperature ranges during competition conditions.
- These requirements have been used to define test cases, guide integration decisions, and track subsystem performance throughout development. Each requirement is verifiable via simulation, lab testing, subsystem benchmarking, or field validation, and has been linked to corresponding Technical Performance Measures where applicable.

Interfaces

The Bender Team’s system architecture integrates multiple complex subsystems that interact through well-defined mechanical, electrical, and software interfaces. These interfaces were identified early in the design process and continuously refined through each review stage (SRR, PDR, and CDR) to ensure modularity, maintainability, and robust system integration. All key interfaces, internal and external, were mapped to relevant hierarchy elements and informed both component selection and physical layout.

Key Interface Types and Descriptions

Interface Type	Description	Affected Subsystems	External/Internal	Key Requirements
Mechanical	Drivetrain assembly interfaces to the chassis through dual driveshafts and aluminum extrusions. Regolith system's rotating drum is actuated via a full-width cross-shaft. Tracks mount modularly for easy field replacement.	1.2.0 Chassis, 1.4.0 Drivetrain, 1.3.0 Regolith Handling	Internal	1.4.3 Terrain Traversal, 1.4.5 Mounting
Electrical	Power is routed through a central bus and subsystem-specific modular harnesses. Dust-proof pass-throughs protect connections between the chassis interior and moving components.	1.2.2 Power Distribution, 1.3.1 Appendage Control, 1.4.1 Power	Internal	1.2.2 E-Stop, 1.3.1 Regolith Power
Signal & Data	Internal Ethernet switch routes navigation data between LiDAR, camera systems, and onboard processing. USB pass-throughs allow communication with debugging interfaces, VR tracking, and WiFi antennas.	1.1.0 Navigation, 1.2.4 Processing & Communication, 1.1.2 Mapping	Internal and External	1.1.2 Mapping, 1.1.1 Localization
Human Interfaces	Physical e-stop mounted externally on the chassis. Lift points are placed based on center-of-mass analysis. Operator interaction via keyboard/mouse or gamepad through a WiFi interface.	1.2.7 Body, 1.2.2 Power Distribution	External	1.2.7 Lifting Points, 1.2.2 E-Stop
Software Interfaces	All subsystems communicate through ROS 2 using shared topics and services, including the navigation stack, mapping nodes, and hardware control layers. Custom drivers unify command and sensor data streams.	1.1.3 Planning, 1.1.4 Control, 1.3.1 Appendage Control	Internal	1.1.4 Compute Time, 1.1.4 Control

Interface Design Considerations and Impacts

All interface considerations were integrated into system design from the earliest concept phases:

- The mechanical interfaces were designed for modularity, allowing subsystems like the drivetrain and regolith arm to be quickly detached and serviced without disturbing

neighboring components. This supports rapid testing and field repair during the competition.

- The electrical layout focused on dust mitigation and serviceability. Centralized power routing reduced cable clutter and simplified failure isolation. Dust-proof pass-throughs protect sensitive electronics in the chassis while allowing flexibility for motion in external components.
- The signal and software interfaces relied on modular communication protocols and ROS2 to manage inter-process communication. This enabled parallel development and testing of subsystems and ensured compatibility during integration.
- External human interfaces, such as the emergency stop and lifting handles, were designed in compliance with international safety standards (IEC 60204-1, ISO 7000-1368) to ensure safe operation and transport.

Interface-Related Requirements

Several key system requirements are interface-driven:

- 1.2.2 – Emergency Stop: Requires both hardware and human interface considerations.
- 1.2.7 – Lifting Points: Directly influenced mechanical layout and body design.
- 1.1.2 – Mapping & 1.1.1 – Positional Accuracy: Dependent on communication interfaces between sensors and processors.
- 1.3.1 – Regolith Power: Drives modular electrical design to isolate and protect the appendage control system.

Through careful planning and control of interfaces across mechanical, electrical, and software domains, the Bender Team ensured a highly integrated, modular design that supports both in-lab testing and real-world deployment.

Engineering Specialties

The Bender Team prioritized several engineering specialties in the development of our autonomous lunar excavation robot to ensure safe, reliable, and adaptable performance throughout the competition lifecycle. These specialties directly shaped design decisions, trade studies, and operational strategies, and were embedded into both the system hierarchy and key requirements.

Maintainability

To support rapid repairs and subsystem replacements during integration and at the competition, we prioritized maintainability across all aspects of the robot:

- Design Influence: The use of aluminum extrusion framing enabled modular mounting for subsystems, allowing any component to be quickly detached without disturbing the rest of the system. Wiring harnesses were labeled and routed along standard paths to streamline debugging and replacement.
- Requirement Influence:
 - 1.2.7 – Lifting Points support maintainability by enabling safe and easy physical access to components.
 - Modular harnessing and labeled connectors support fast electrical diagnostics and repairs.
- Trade Studies: Maintainability was a major consideration in selecting aluminum framing over welded tubing and in opting for modular rather than integrated PCB designs. Maintainability was also considered when deciding on and designing 3d-printed tracks

Transportability

Given the need to ship the robot across long distances and reassemble it under time constraints, transportability was a key concern:

- Design Influence: The robot was designed to break into four major components: chassis, drivetrain, regolith system, and navigation module. Fasteners and connectors were selected for tool compatibility and ease of access.
- Requirement Influence:
 - 1.0.0 – System Dimensions ensured compliance with transport and competition constraints.
 - 1.2.7 – Lifting Points also enabled safe lifting and secure packing during transport.
- Trade Studies: Transportability factored into the decision to use 3D-printed components for the drivetrain, reducing weight and bulk for shipping, and into the overall system architecture layout.

Safety

Protecting team members, reviewers, and competition staff was an essential design priority:

- Design Influence:
 - Integrated physical and software emergency stops, placed in accessible locations.
 - The system was designed to allow remote startup and testing to minimize close-contact interaction during autonomous operation.
 - Sharp edges were rounded off and pinch points were minimized across the mechanical assembly.
- Requirement Influence:

- 1.2.2 – E-stop ensures a rapid shutdown path compliant with OSHA and IEC standards.
- 1.2.7 – Lifting Points help prevent overloading or unbalanced lifts.
- Trade Studies: Safety influenced the decision to use commercial off-the-shelf e-stop components and informed layout decisions around the control panel.

Reliability

The system was engineered for robustness and consistent performance across all test and competition environments:

- Design Influence:
 - Simplicity and accessibility guided subsystem layout to reduce failure points.
 - Redundant sensor systems (e.g., LiDAR + stereo vision + IMU) were used to ensure continued localization in the event of partial sensor failure.
 - Passive cooling and dust-sealed components help protect electronics during extended operation.
- Requirement Influence:
 - 1.4.3 – Terrain Traversal and 1.3.0 – Regolith Management depend on reliable drivetrain and regolith system operation even under varied load and terrain conditions.
 - 1.1.4 – Control and 1.1.1 – Localization depends on reliable system timing and sensor fusion.
- Trade Studies: Reliability was a key criterion in material selection, sensor choice, and the decision to use a track system over wheels.

Summary

These engineering specialties—maintainability, transportability, safety, and reliability—were not only critical to mission success but also deeply integrated into our requirements, component choices, and trade studies. They helped reduce project risk, improve field readiness, and ensure that Bender can be operated, shipped, and maintained effectively throughout the competition lifecycle.

Concept of Operations

The Bender Team’s lunar excavation robot is designed to operate primarily under fully autonomous control, with a robust manual fallback mode in the event of system failure. The robot’s operational plan is structured to maximize excavation efficiency while ensuring reliability and safety within the constraints of the simulated lunar environment.

Startup and Initialization

Before each mission, the robot is powered on using a physical e-stop switch. Once the onboard system boots, a command-line initialization script is executed to configure the system and bring up the required services.

Subsystems are initialized in the following sequence:

1. Navigation stack
2. Network and ROS communication nodes
3. Motor calibration routines
4. Remaining subsystems that do not require calibration (e.g., regolith control logic, logging services)

This structured initialization ensures that positional data is accurate and control systems are fully aligned before autonomous operations begin.

Mission Execution

The autonomous sequence is triggered by a remote signal from the operator. Once activated, the robot performs the following steps:

1. Initial Terrain Scan: A loop of the arena is performed using a custom SLAM algorithm to generate a high-resolution map of the environment. Points outside the arena boundaries are ignored to prevent use of the walls in planning and localization.
2. Obstacle Avoidance and Navigation: The robot plans a path to the excavation zone using data from multiple LiDARs, a stereoscopic camera, an IMU, and absolute position data from a Vive beacon tracker. Obstacle data is used to filter viable dig sites from inaccessible areas.
3. Regolith Collection: Once positioned, the bucket drum begins excavation. After collecting the regolith, the robot raises the arm and proceeds to the dump zone.
4. Dynamic Dumping: The robot evaluates the current regolith mound height and selects a dynamically determined dump location. The drum spins in reverse to release material in a controlled manner.
5. Cycle Continuation: The robot repeats the collection/dump cycle until time runs out or a manual stop is triggered.

Mission Termination

The mission may be stopped in one of two ways:

- Soft stop: A remote signal pauses autonomous operation and resets the vehicle to a resting state.

- Hard stop: The emergency stop switch immediately powers down the system. Recovery requires a full power cycle and reinitialization.

There is no "home" return behavior, however, the robot can be sent to a safe resting position via remote signal once operations are complete.

Manual Fallback Mode

In the event of autonomy failure, the robot can be switched to manual control using a wireless gamepad. This allows the operator to:

- Drive the robot manually
- Control the drum for collection or dumping
- Monitor environment data through streamed point cloud visuals (no need for live video feed)

There are no functional limitations in fallback mode, all operations can be completed manually.

Environmental Considerations

The robot is designed to operate in dusty, uneven terrain with intermittent sensor blind spots, most notably a known beacon occlusion zone behind the central pillar in the simulated lunar environment. The system is engineered to rely on a combination of inertial sensing and local mapping to maintain stability in these regions.

Operations-Driven Requirements

The following key requirements were specifically shaped by operational considerations:

- Autonomous Control: The robot shall perform full excavation and dumping cycles without human input (linked to mission objectives and SLAM navigation).
- Regolith Collection Efficiency: Excavation shall be performed using a rotating drum, with the amount of collected material normalized to power consumption and system weight.
- Obstacle Detection and Navigation: The robot shall detect and avoid obstacles using a fused sensor network with real-time decision-making.
- Modular Manual Control: All autonomous functions shall be executable in manual mode without limitations.

Technical Performance Measurement

Throughout development, Team Bender used a set of carefully defined Technical Performance Measures (TPMs) to evaluate system design progress, mitigate technical risk, and ensure mission success. These TPMs served as quantifiable checkpoints for critical subsystems and were reviewed at each major design milestone (SRR, PDR, and CDR). The TPMs were allocated across the system hierarchy and adjusted based on testing outcomes, trade studies, and reviewer feedback.

Most Critical TPMs

TPM	Target	Importance
Positional Accuracy	± 5 cm	Enables accurate navigation and mission execution
Mapping Accuracy	± 5 cm	Critical for detecting obstacles and valid dig/dumpsites
Compute Time	≤ 500 ms	Maintains real-time responsiveness in the autonomy loop
Terrain Traversal	Transport 10 kg regolith successfully	Mission-critical for scoring and functionality
Drivetrain Weight	≤ 15 kg	Impacts chassis stability, total mass, and traction
Chassis Weight	≤ 30 kg	Supports lifting, safety, and transportability
Regolith Power Use	≤ 200 W	Protects power budget and ensures safe operation

TPM Allocation by System Hierarchy

System Element	TPMs Allocated
1.1 – Navigation	Positional Accuracy, Mapping Accuracy, Compute Time
1.2 – Chassis	Chassis Weight, E-stop accessibility, Lifting Point Safety
1.3 – Regolith Handling	Power Limit, Handling Weight, Dump Accuracy
1.4 – Drivetrain	Power Use, Weight, Turning Radius, Terrain Traversal

Evolution of TPMs During Development

- **Positional Accuracy and Mapping:** These were initially set with generous bounds and refined to ± 5 cm after early sensor fusion testing showed acceptable drift margins using the Vive beacon and IMU.
- **Compute Time:** Originally budgeted for ≤ 750 ms, this was reduced to ≤ 500 ms after transitioning to a multi-threaded ROS2 framework and optimizing map generation speed.
- **Drivetrain Weight and Traction:** Adjusted after the PDR due to the switch from wheels to 3D-printed tracks, which increased mechanical complexity but improved terrain

performance. A TPM for track durability was considered but folded into field verification testing.

- **Regolith Power and Handling:** Increased slightly during testing due to changes in motor selection and torque demands of the drum. Power budgeting helped maintain the cap at 200W with careful load distribution.
- **Chassis Weight:** The original design involved a front and back declined top panel to minimize regolith accumulation. This was reduced to a single decline to increase internal room for a motor. As more components were added, the electronics layout was adjusted for even weight distribution for the center of mass and regolith handles.

TPMs as a Design and Management Tool

TPMs were embedded into the decision-making process across the project:

- Used in trade studies to weigh performance vs. risk/cost.
- Monitored during integration to flag underperforming components early.
- Served as control gates for reviews, where design readiness was based on TPM compliance.
- Tied directly to testing procedures and final field validation.

This structured use of TPMs helped ensure that critical elements of the system met or exceeded competition performance thresholds while minimizing late-stage risk.

Trade Studies

To guide critical design decisions, Team Bender conducted a series of trade studies throughout the early development phases, particularly during the SRR and PDR milestones. These studies involved the comparison of design alternatives using weighted evaluation matrices that considered technical feasibility, system complexity, manufacturability, cost, mass, and alignment with mission objectives.

Each study was scored collaboratively by subteams and leads, using a consistent rubric. The top-performing solutions were selected based on total weighted score and alignment with competition requirements. The results of these trade studies directly shaped several key driving requirements in our system design.

Tracks vs. Wheels

Context: Mobility system selection for effective terrain traversal and regolith transport.

Criteria Considered:

- Traction and surface area
- Weight

- Power consumption
- System complexity
- Manufacturability
- Risk of getting stuck

Result: Tracks were selected due to higher terrain adaptability, very low turning radius, and lower ground pressure, even though they introduced slightly higher complexity and maintenance needs.

Resulting Requirement:

- 1.4.3 – Terrain Traversal: “The vehicle shall be able to traverse the simulated lunar environment while transporting 10 kg of regolith without losing locomotive capability.”
- 1.4.0 – Drivetrain Weight: Kept under 15 kg despite switching to tracks.

Aluminum Frame vs. Welded Steel or Composites

Context: Chassis material selection for strength, manufacturability, and maintainability.

Criteria Considered:

- Weight
- Ease of fabrication
- Modifiability
- Structural stiffness
- Cost

Result: Aluminum extrusion was chosen for its balance of low weight, ease of assembly, and ability to accommodate modular mounts and changes throughout development.

Resulting Requirement:

- 1.2.0 – Chassis Weight: “The chassis of the vehicle shall have a mass of 30 kg or less.”

Regolith Excavation Method: Drum vs. Bucket Arm vs. Auger

Context: Selection of regolith collection mechanism for efficient operation with minimal dust disturbance.

Criteria Considered:

- Collection efficiency
- Dust mitigation
- Simplicity
- Power draw
- Durability

Result: The rotating drum was selected for its continuous collection capability and lower dust disturbance during excavation and dumping.

Resulting Requirements:

- 1.3.0 – Regolith Handling: Full coverage of mass, deposit accuracy, and collection strategy.
- 1.3.1 – Regolith Power: “The regolith subsystem shall not exceed 200 Watts.”

Sensor Suite for Navigation

Context: Selection of sensors for obstacle detection, localization, and autonomous control.

Criteria Considered:

- Accuracy
- Range
- Computational load
- Sensor fusion compatibility
- Cost

Result: A combined system of multiple LiDARs, a stereoscopic camera, an IMU, and a Vive beacon was chosen to provide both global and local localization, enabling robustness in blind zones.

Resulting Requirements:

- 1.1.1 – Positional Accuracy: “The navigation system shall locate position with an accuracy of ± 5 cm.”
- 1.1.2 – Mapping: “The vehicle shall detect obstacles with accuracy within ± 5 cm.”
- 1.1.4 – Compute Time: Capped at 500 ms to maintain real-time control.

Manual vs. Fully Autonomous Operation

Context: Deciding how the vehicle should operate during the competition and in fallback scenarios.

Criteria Considered:

- Mission score potential
- Risk of autonomous failure
- Field adaptability
- Operator workload

Result: The team chose to design for full autonomy, while also building a manual fallback system via gamepad control and streamed point cloud feedback.

Resulting Requirements:

- Autonomy Level Requirement (Operational): Full mission execution without human intervention.
- Fallback Control Capability: All autonomous functions must be fully operable manually with no limitations.

Summary

Trade studies played a critical role in shaping the technical foundation of Team Bender's design. The decisions made through structured evaluations of alternatives not only helped reduce risk but also directly informed several of our key driving requirements, ensuring that each subsystem met mission objectives while balancing complexity, cost, and field readiness.

Verification of System Meeting Requirements

Team Bender employed a multi-stage verification strategy to ensure that the final robot system satisfies all key driving requirements under both expected environmental conditions and realistic operational workflows. Our approach combined simulation, lab-based subsystem testing, system-level dry runs, and full-environment field testing to validate performance against technical metrics and operational goals.

Verification was integrated into the development schedule following each major review milestone (SRR, PDR, and CDR) and tracked through checklists and documented test procedures. Final integration testing is ongoing and will culminate in full-system testing at the Bruneau Sand Dunes, simulating the lunar competition terrain.

Verification Methods by Requirement Category

Requirement	Verification Method	Success Criterion
1.1.1 – Positional Accuracy	SLAM simulation, indoor beacon tests	Robot localizes within ± 5 cm of ground-truth marker
1.1.2 – Mapping	Obstacle field tests using known targets	Obstacle boundaries detected with ± 5 cm accuracy
1.1.4 – Compute Time	ROS timing logs, runtime analysis	SLAM and planning run in < 500 ms per cycle
1.2.0 – Chassis Weight	Digital scale measurement	Total chassis < 30 kg
1.2.2 – Emergency Stop	Field and bench testing	Manual and digital e-stops fully cut power on command
1.2.7 – Lifting Points	Load simulation and manual lift	No single point exceeds 20 kg; lifting safe and balanced
1.3.0 – Regolith Handling	Sandbed testing with drum actuation	Collects, holds, and deposits material accurately
1.3.1 – Regolith Power	Inline power logging during the operation	≤ 200 Watts used during peak excavation

1.4.0 – Drivetrain Weight	Measured with full track system installed	≤15 kg total drivetrain mass
1.4.1 – Drivetrain Power	Power draw monitored during traction trials	≤400 Watts under full load
1.4.2 – Turning Performance	Indoor mock arena test	The robot performs 180° turn within a 2m radius
1.4.3 – Terrain Traversal	Simulated regolith bed transport tests	Successfully traverses terrain with a 10 kg load
1.0.0 – Vehicle Dimensions	Manual measurement + 3D CAD check	Fits within 1.5m × 0.75m × 0.75m envelope
Operational Autonomy	Field testing with live mapping and regolith dump	Completes dig-dump cycle with no manual input

Verification of Interfaces

Key interfaces were verified at both subsystem and system integration levels:

- Mechanical Interfaces: Verified through dry fits and stress testing (e.g., track attachment, arm pivot shaft, chassis mounts). All mounting systems were torque-tested and adjusted for rigidity.
- Electrical Interfaces: Power harnesses were continuity-tested and thermally scanned during operation. Dust-proof pass-throughs were evaluated in mock dusty environments to verify seal integrity and stable operation.
- Signal & Software Interfaces: Verified using ROS tools such as rqt_graph, ros2 topic echo, and timed loopbacks to ensure latency remained below threshold across Ethernet and USB pass-throughs.
- Human Interfaces: E-stop functionality was tested under load and during simulated faults. Lifting points were tested for balance using dead weights.

Field Testing Plan

Final system validation will be conducted in two environments:

- University-constructed sandbed (controlled environment): Used to verify all autonomy behaviors, drivetrain reliability, regolith management cycles, and safety procedures.
- Bruneau Sand Dunes (natural terrain): Used for stress-testing traversal, mapping, and full system integration under field conditions to generate our required “proof of life” submission.

Traceability and Success Criteria

Each verification test has been mapped back to a requirement and includes:

- A defined test objective
- A pass/fail metric
- Data logging procedures
- Contingency plans for failure cases

All success criteria are directly derived from competition rules or internal TPM thresholds, ensuring traceable and relevant results for competition readiness.

Conclusion

Through the structured application of systems engineering principles, Team Bender successfully developed a fully autonomous lunar excavation robot from a clean-sheet design. By leveraging rigorous requirements definition, modular system hierarchy, formal design reviews, and iterative verification, we created a robust and adaptable platform capable of navigating and performing in the simulated lunar environment. The engineering decisions were guided by mission-critical trade studies, thoughtful interface management, and real-world constraints, ensuring both technical excellence and operational readiness. As first-time competitors, this process not only shaped a competitive design but also established a scalable engineering workflow for future missions and teams to build upon.

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